Word 2 Vec → Feature Representation

Word2Vec is a technique for natural language processing published in 2018. The word2vec algorithm uses a neural network model to learn word associations from a large corpus of text. Once trained, such a model can detect synonymous words or suggest additional words for a practial sentence. As the name implies, word2vec

{Note:} Word2Vec is develop by Google

Vocabulary → Unique words → corpus

		Boy	Girl	KING	QUEEN	Apple	Mango
Feature Representation	Gender	-1	1	-0.92	+0.93	0.01	0.05
	Royal	0.01	0.02	0.95	0.96	-0.02	0.02
	Age	0.03	0.04	0.75	0.68	0.95	0.96
	food	-	-	-	-	0.91	0.92
	⋮	-	-	-	-	-	-
300 dimension	n^{th}	-	-	-	-	-	-

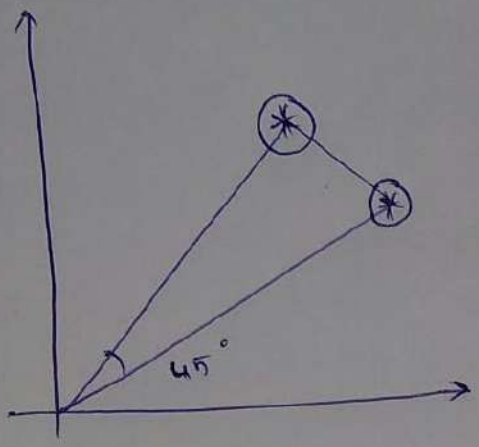
[KING - BOY + QUEEN = GIRL]

KING [0.95 , 0.96] MAN [0.95 , 0.98]

QUEEN [-0.96 , 0.95] women [-0.94 , -0.96]

KING - MAN + QUEEN = WOMEN

Cosine similarity



Distance = 1 - Cosine similarity

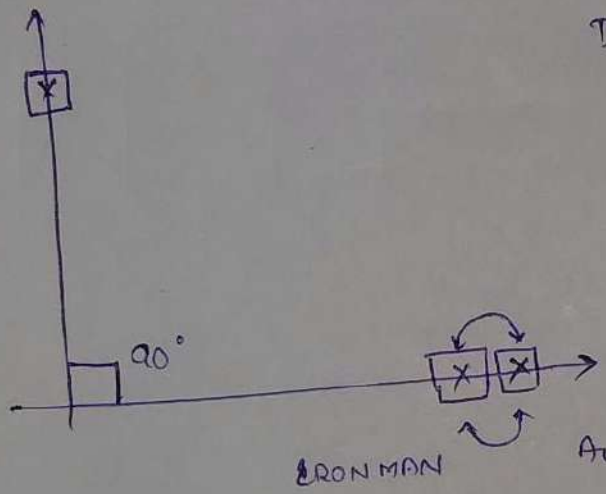
Cosine - Sim = $\cos 45 = \frac{1}{\sqrt{2}}$

$= 0.7071$

Distance = $1 - 0.7071$

$\longrightarrow = 0.29$

$\boxed{1-1=0}$



Distance = $1 - \cos\theta$

$= 1 - \cos 90$

$= 1 - 0$

$= 1$

Distance = $1 - \cos\theta$

$= 1 - 1$

$= 0$

